

BCA 1st year LOC Question Bank

Unit-1

- 1) Write the binary coding for the word BOY in:
 - a. ASCII-8.
 - b. EBCDIC
- 2) Represent the decimal number 8620 in:
 - a.) BCD
 - b.) Excess-3 code
 - c.) 2421 code
 - d.) As a binary number
 - e.) Explain floating point representation
 - f.) Make half adder for adding two numbers
 - g.) Use 2's complement to solve
 - i. $-9 - 10$
 - ii. $-32 + 16$
 - iii. $-7 - 19$
 - iv. $+7 + 6$
 - v. $-9 - 7$
 - vi. $-32 + 10$
 - vii. $-6 - 9$
 - viii. $(1011)_2 - (101)_2$
- 3) A. Solve the following complements Representation of numbers by using Suitable Example:
 - i. True Complement:
 - ii. Radix-Minus-One Complement.A. Solve the following;
 - A. $(1101.1)_2 * (111.01)_2 = (?)_2$
 - B. $(11000)_2 - (11)_2 = (?)_2$
 - C. $(130)_{10} = ()_3$
 - D. $(1000)_3 = ()_{10}$
 - E. $(8554)_{10} = ()_6$
 - F. $(17.3)_{10} = ()_2, ()_{16}$
 - G. $(234)_{10} = ()_5$
 - H. $(221)_6 = ()_{10}$
 - I. $(?)_2 = 3AB7$
 - J. $(109)_{16} = ()_2$
 - K. $(101101101.101)_2 = ()_8$
 - L. $(125.125)_{10} = ()_2$
 - M. $(11.625)_{10} = ()_2, ()_8, ()_{16}$
 - N. $(110111101)_2 = ()_8, ()_{16}$
 - O. Register stores high low, high, low .find number in decimal, octal and hexa decimal.
 - P. Write a note on 7-bit representation for error detection and correction code.
 - Q. Abbreviate ASCII, EBCDIC
 - R. Draw and label 3 variable k-map

- S. Explain normalized representation of floating point numbers.
- T. What is parity bit? define odd and even parity
- U. State duality principle
- V. Make T.T for self complementing code
- W. What is ones complement

4) i. give comparison between 1's and 2's complement

- B. Why do digital computers use binary number system?
- C. What do you mean by fixed point representation of numbers?
- D. Differentiate between coding and convention?
- E. Define cyclic code and 8421 code
- F. What is condition of overflow and underflow in floating point notation

5.) Perform the following using 8-bit notation and 2's complement

i) $(78)_{10} - (36)_{10}$

ii) $-(45)_{10} - (35)_{10}$

6.) write method of subtraction using one's complement give example.

7. explain error detecting and correcting code give example